Linear Algebra and Its Applications

Section 5.S: Supplementary Exercises

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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5.S Chapter 5 supplementary exercises

These exercises connect eigenvectors, similarity, polynomial expressions in matrices, and diagonalization. Keep three distinctions in view: row equivalence does not preserve eigenvalues; similarity does. An eigenvalue may be zero, but an eigenvector must be nonzero. A repeated eigenvalue does not by itself decide diagonalizability: the total number of independent eigenvectors decides it. All matrices are real unless complex roots or complex diagonalization are explicitly being considered.

Exercise 5.S.1: Twenty-four statements with proofs or counterexamples

(a) **True.** If $Ax = x$ and $x \neq 0$, multiplication by A^{-1} gives $A^{-1}x = x$.

(b) **False.** $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ is invertible and row equivalent to I, but has only one independent eigenvector, since its 1-eigenspace is $\text{Span}\{(1,0)^T\}$.

(c) **True.** A zero column makes $Ae_j = 0$ for some nonzero coordinate vector. A zero row makes the matrix singular as well; by the Invertible Matrix Theorem its kernel contains a nonzero vector.

(d) **False.** The scalar matrix [2] has eigenvalue 2, whereas its square [4] has only eigenvalue 4.

(e) **True.** $Ax = \lambda x$ implies $A^2x = \lambda^2 x$, with the same nonzero x.

(f) **True.** An invertible A has $\lambda \neq 0$; $Ax = \lambda x$ gives $A^{-1}x = \lambda^{-1}x$.

(g) **False.** The zero matrix has eigenvalue zero.

(h) **True.** Nonzeroness is part of the eigenvector definition; otherwise $A0 = \lambda 0$ would make every scalar an eigenvalue.

(i) **False.** For I_2, the independent vectors e_1, e_2 have the same eigenvalue 1.

(j) **True.** Similarity gives $\det(P^{-1}(A - \lambda I)P) = \det(A - \lambda I)$, so the characteristic polynomials, including multiplicities, agree.

(k) **False.** For $A = \text{diag}(1, 2)$ and $B = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$ the eigenvalues agree and the matrices are similar (both have independent eigenbases), but e_2 is an eigenvector of A and not of B.

(l) **False.** For $A = \text{diag}(1, 2)$, $e_1 + e_2$ is not an eigenvector. Even within one eigenspace, the sum of v and $-v$ is zero and hence not an eigenvector.

(m) **False.** All diagonal entries of an upper triangular matrix are eigenvalues, including zeros.

(n) **True.** $\det(A^T - \lambda I) = \det((A - \lambda I)^T) = \det(A - \lambda I)$.

(o) **False.** I_5 has only one distinct eigenvalue but is already diagonal; its five coordinate vectors form an eigenbasis.

(p) **True.** The real rotation matrix $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ has eigenvalues $\pm i$ and no real eigenvectors.

(q) **False.** $\text{diag}(0, 1)$ is diagonalizable but its first column is zero, so its columns are dependent.

(r) **True.** If $Av = \lambda v = \mu v$ and $v \neq 0$, then $(\lambda - \mu)v = 0$ forces $\lambda = \mu$.

(s) **False.** Invertibility and diagonalizability are different. The non-diagonalizable invertible example in (b) refutes one direction; $\mathrm{diag}(0, 1)$ refutes the reverse direction.

(t) **True.** If $Ae_j = \lambda_j e_j$, column j has only its diagonal entry possibly nonzero, so A is diagonal.

(u) **True.** If $A = SBS^{-1}$ and $B = PDP^{-1}$, then $A = (SP)D(SP)^{-1}$, a diagonalization.

(v) **True.** $A^{-1}(AB)A = BA$, so the products are similar. In fact invertibility of A alone suffices.

(w) **False.** The zero matrix has n independent eigenvectors but is not invertible.

(x) **True.** Diagonalizability supplies an eigenvector basis, and the definition of a basis says every vector is a linear combination of its vectors (over $\mathbb{R}$ for real diagonalization here).

Exercise 5.S.2: Move an eigenvector through the right-hand factor

Suppose $(AB)x = \lambda x$ and $Bx \neq 0$. Associativity gives

$$(BA)(Bx) = B(ABx) = B(\lambda x) = \lambda(Bx).$$

Because Bx is nonzero by assumption, it is an eigenvector of BA for the same eigenvalue. The condition $Bx \neq 0$ is what prevents the zero vector from being incorrectly called an eigenvector.

Exercise 5.S.3: Apply two polynomials to an eigenvector

(a) $(5I - A)x = 5x - Ax = (5 - \lambda)x$. Thus x remains a nonzero eigenvector, with eigenvalue $5 - \lambda$.

(b) First $A^2x = A(\lambda x) = \lambda^2 x$. Consequently $(5I - 3A + A^2)x = (5 - 3\lambda + \lambda^2)x$. The new eigenvalue is $5 - 3\lambda + \lambda^2$.

Exercise 5.S.4: Induct on the matrix power

For $m = 1$ the assertion is the assumed eigenvector equation. Suppose $A^m x = \lambda^m x$. Multiplying by A gives $A^{m+1}x = A(\lambda^m x) = \lambda^m Ax = \lambda^{m+1}x$. The same x remains nonzero at every stage, so each displayed scalar is genuinely an eigenvalue of the corresponding power. This proves the claim for every positive integer m.

Exercise 5.S.5: Polynomial evaluation preserves eigenvectors

Exercise 5.S.4 gives $A^j x = \lambda^j x$ for every positive power; for $j = 0$ the same equality is $Ix = x$. Therefore

$$p(A)x = \sum_{j=0}^{n} c_j A^j x = \sum_{j=0}^{n} c_j \lambda^j x = p(\lambda)x.$$

Since $x \neq 0$, $p(\lambda)$ is an eigenvalue of $p(A)$. Different eigenvalues of A can map to the same polynomial value; that does not invalidate the conclusion.

Exercise 5.S.6: Diagonalize polynomial expressions using the original basis

(a) With $A = PDP^{-1}$, neighboring factors $P^{-1}P$ cancel in A^2. Since $I = PIP^{-1}$, $5I - 3A + A^2 = P(5I - 3D + D^2)P^{-1}$. At diagonal entries $2, 7$, the polynomial has values $3, 33$. Thus $B = P\,\mathrm{diag}(3, 33)P^{-1}$ is a diagonalization.

(b) The same cancellation gives $A^j = PD^jP^{-1}$ for every power. Adding the finitely many terms gives $p(A) = Pp(D)P^{-1}$, with diagonal entries $p(2), p(7)$. This remains a diagonalization even if those two values coincide.

Exercise 5.S.7: Cayley–Hamilton for a diagonalizable matrix

Write $A = PDP^{-1}$. Every diagonal entry λ_j of D is an eigenvalue of A and hence a root of its characteristic polynomial p. By the polynomial identity in Exercise 5.S.6,

$$p(A) = Pp(D)P^{-1} = P\,\mathrm{diag}(p(\lambda_1), \ldots, p(\lambda_n))P^{-1} = P0P^{-1} = 0.$$

This proves the theorem under the stated diagonalizability assumption; it does not pretend that every square matrix is already known to be diagonalizable.

Exercise 5.S.8: One eigenvalue plus diagonalizability forces a scalar matrix

(a) If the only eigenvalue is λ with multiplicity n, a diagonalization has $D = \lambda I$. Therefore $A = P(\lambda I)P^{-1} = \lambda PP^{-1} = \lambda I$.

(b) The given $\begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ has only eigenvalue 3, twice, but is not $3I$. By (a), it cannot be diagonalizable. Directly, $(A - 3I)x = 0$ forces $x_2 = 0$, leaving only one independent eigenvector.

Exercise 5.S.9: An eigenvalue inside the unit disk cannot equal one

If $I - A$ were singular, a nonzero v would satisfy $(I - A)v = 0$. Then $Av = v$, so 1 would be an eigenvalue of A. Its modulus is 1, contradicting the strict bound on every eigenvalue. Hence $I - A$ is invertible. This argument requires no diagonalizability assumption.

Exercise 5.S.10: Every eigenmode decays and therefore the whole matrix does

Use $A = PDP^{-1}$. Its powers satisfy $A^k = PD^kP^{-1}$. Every diagonal entry of D^k is $\lambda_j^k \to 0$ because $|\lambda_j| < 1$. Thus each entry of PD^kP^{-1} is a finite sum of fixed constants times quantities tending to zero, so it also tends to zero. Equivalently, apply the mode decomposition to each coordinate vector e_j; then $A^k e_j \to 0$, which says each column of A^k tends to zero.

Conclusion. $A^k \to 0$ entry by entry.

Exercise 5.S.11: Invariant lines and real eigenvectors

(a) Every $x \in H$ has the form $x = cu$. Then $Ax = cAu = c\lambda u \in H$. This is exactly invariance.

(b) Choose a nonzero u spanning the one-dimensional invariant subspace K. Invariance gives $Au \in K$, so $Au = \lambda u$ for some scalar λ. Since $u \neq 0$, it is an eigenvector.

Exercise 5.S.12: Factor a block triangular characteristic polynomial

In the permutation expansion of $G = \begin{bmatrix} A & X \\ 0 & B \end{bmatrix}$, a nonzero term must assign the bottom block rows to bottom block columns; otherwise it uses a zero from the lower-left block. The remaining top rows use top block columns. The products and signs therefore separate into the determinant sums for A and B, giving $\det G = \det A \det B$. The same argument applied to $G - \lambda I = \begin{bmatrix} A - \lambda I & X \\ 0 & B - \lambda I \end{bmatrix}$ gives the product of the two characteristic polynomials. No inverse of either diagonal block is required.

Exercise 5.S.13: Read one eigenvalue and solve the remaining block

The upper-left block is $[3]$ and the lower-right block is $B = \begin{bmatrix} 5 & -2 \\ -4 & 3 \end{bmatrix}$. Its characteristic polynomial is $(\lambda - 5)(\lambda - 3) - 8 = \lambda^2 - 8\lambda + 7 = (\lambda - 1)(\lambda - 7)$. By Exercise 5.S.12 the complete eigenvalue list is $3, 1, 7$, each with multiplicity one. The entries in the upper-right block have no effect on this list.

Exercise 5.S.14: Combine the spectra of two diagonal blocks

The blocks are $B = \begin{bmatrix} 1 & 5 \\ 2 & 4 \end{bmatrix}$ and $C = \begin{bmatrix} -7 & -4 \\ 3 & 1 \end{bmatrix}$. Their characteristic polynomials are

$$\det(\lambda I - B) = \lambda^2 - 5\lambda - 6 = (\lambda - 6)(\lambda + 1),$$

$$\det(\lambda I - C) = \lambda^2 + 6\lambda + 5 = (\lambda + 1)(\lambda + 5).$$

Multiply them by the block result. The eigenvalues are $6, -5$, each once, and -1 twice. Algebraic multiplicity is counted even when an eigenvalue occurs in both blocks.

Exercise 5.S.15: Split into the constant vector and coordinate-sum-zero space

Let $\mathbf{1} = (1, \ldots, 1)^T$. Then $J\mathbf{1} = n\mathbf{1}$, so $A\mathbf{1} = [a + (n-1)b]\mathbf{1}$. For z with coordinate sum zero, $Jz = 0$, and hence $Az = (a - b)z$. The coordinate-sum-zero subspace has dimension $n - 1$; together with $\mathbf{1}$ it gives a basis of $\mathbb{R}^n$. Thus the characteristic polynomial is

$$(\lambda - (a - b))^{n-1}(\lambda - (a + (n-1)b)).$$

When $b \neq 0$ and $n \geq 2$, the two eigenvalues are distinct, with multiplicities $n - 1$ and 1. If $b = 0$, they coincide at a and the combined multiplicity is n; for $n = 1$ only the value a is present. These qualifications prevent double-counting a coincident value.

Exercise 5.S.16: Apply the common-diagonal formula twice

For the 3×3 matrix, $a = 1, b = 2$. Thus $a - b = -1$ has multiplicity 2, and $a + 2b = 5$ has multiplicity 1. For the 5×5 matrix, $a = 7, b = 3$. Thus $a - b = 4$ has multiplicity 4, and $a + 4b = 19$ has multiplicity 1. These multiplicities add to the dimension in each case.

Exercise 5.S.17: The trace and determinant decide reality in dimension two

For a real matrix $A = [a_{ij}]$, expand

$$\det(\lambda I - A) = (\lambda - a_{11})(\lambda - a_{22}) - a_{12}a_{21} = \lambda^2 - (\operatorname{tr} A)\lambda + \det A.$$

A real quadratic has both roots real exactly when its discriminant is nonnegative. Here that condition is $(\operatorname{tr} A)^2 - 4 \det A \geq 0$, equivalently

$$\det A \leq \left(\frac{\operatorname{tr} A}{2}\right)^2.$$

Equality includes a repeated real eigenvalue; it does not by itself assert diagonalizability.

Exercise 5.S.18: Compute the limiting spectral projection

The matrix has eigenvalues 1 and $3/5$, with eigenvectors $(-1, 2)^T$ and $(-3, 2)^T$. Distinct eigenvalues give a basis. The projection onto the 1-eigenspace along the other eigenspace is

$$E = \frac{A - (3/5)I}{1 - 3/5} = \begin{bmatrix} -1/2 & -3/4 \\ 1 & 3/2 \end{bmatrix}.$$

Indeed E acts as identity on the first eigenvector and as zero on the second, so on the whole eigenbasis

$$A^k = E + (3/5)^k(I - E).$$

The second term tends to zero. Therefore A^k tends to the stated matrix E, whose negative entries cause no contradiction: the original A already has a negative off-diagonal entry.

Exercise 5.S.19: A quadratic companion matrix

For $p(t) = 6 - 5t + t^2$, the companion matrix in the book's convention is

$$C_p = \begin{bmatrix} 0 & 1 \\ -6 & 5 \end{bmatrix}.$$

Its characteristic polynomial is $\det(\lambda I - C_p) = \det \begin{bmatrix} \lambda & -1 \\ 6 & \lambda - 5 \end{bmatrix} = \lambda^2 - 5\lambda + 6 = p(\lambda)$. If using $\det(C_p - \lambda I)$ instead, the sign is the same in even dimension.

Exercise 5.S.20: A cubic companion matrix and its determinant

The coefficients $(-24, 26, -9, 1)$ give

$$C_p = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 24 & -26 & 9 \end{bmatrix}.$$

Expand $\det(\lambda I - C_p)$ along the first row:

$$\lambda \det \begin{bmatrix} \lambda & -1 \\ 26 & \lambda - 9 \end{bmatrix} - 24 = \lambda^3 - 9\lambda^2 + 26\lambda - 24.$$

This equals $(\lambda - 2)(\lambda - 3)(\lambda - 4) = p(\lambda)$. For the book's $\det(C_p - \lambda I)$ convention the polynomial is its negative, as the dimension is odd.

Exercise 5.S.21: Inductively compute the companion characteristic polynomial

For $n = 2$, direct expansion gives $\det \begin{bmatrix} -\lambda & 1 \\ -a_0 & -a_1 - \lambda \end{bmatrix} = \lambda^2 + a_1\lambda + a_0 = (-1)^2 p(\lambda)$. Suppose the identity holds in dimension $n - 1$. Expand $C_p - \lambda I$ down its first column. Only the top and bottom entries can contribute. The top contribution is $-\lambda$ times the determinant of the companion-form block associated with $q(t) = a_1 + a_2 t + \cdots + a_{n-1}t^{n-2} + t^{n-1}$. By induction this determinant is $(-1)^{n-1}q(\lambda)$. The bottom contribution is $(-a_0)(-1)^{n+1}$ times a determinant of 1; the latter minor is triangular with diagonal ones. Thus the total is

$$(-\lambda)(-1)^{n-1}q(\lambda) + (-1)^n a_0 = (-1)^n[\lambda q(\lambda) + a_0] = (-1)^n p(\lambda).$$

This completes the induction for all $n \geq 2$.

Exercise 5.S.22: A polynomial root supplies an explicit companion eigenvector

(a) The companion matrix is $C_p = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix}$.

(b) Since $p(\lambda) = 0$, $\lambda^3 = -a_0 - a_1\lambda - a_2\lambda^2$. Therefore

$$C_p(1, \lambda, \lambda^2)^T = (\lambda, \lambda^2, -a_0 - a_1\lambda - a_2\lambda^2)^T = (\lambda, \lambda^2, \lambda^3)^T = \lambda(1, \lambda, \lambda^2)^T.$$

The first coordinate is 1, so the vector is nonzero even if $\lambda = 0$; it is genuinely an eigenvector.

Exercise 5.S.23: A Vandermonde matrix is an eigenvector matrix

Each column of V is the eigenvector from Exercise 5.S.22 for one root λ_j. Distinct roots give distinct eigenvalues, so the corresponding eigenvectors are independent. Thus V is invertible without needing to compute its inverse. The three eigenvector equations combine into $C_p V = V \operatorname{diag}(\lambda_1, \lambda_2, \lambda_3)$. Multiply on the left by V^{-1} to obtain the requested diagonal matrix. Complex roots are allowed if matrices and independence are interpreted over $\mathbb{C}$.

Exercise 5.S.24: The companion-matrix idea behind a polynomial-root command

The basic method documented for MATLAB's `roots` is to construct a companion matrix and compute its eigenvalues. After removing leading zero coefficients and normalizing the leading nonzero coefficient, the polynomial is monic. Exercise 5.S.21 shows that the companion matrix's eigenvalues are exactly its roots. Numerical eigenvalue algorithms, typically using QR-based methods, then approximate those values. The companion matrix used by a program may be transposed or written in a different convention; its characteristic polynomial still supplies the same roots. The MATLAB reference is `https://www.mathworks.com/help/matlab/ref/roots.html`. This explains the algorithmic idea, not an assertion that every computed root has exact arithmetic accuracy: companion matrices can be sensitive to coefficient rounding, especially near repeated roots.

Exercise 5.S.25: A repeated eigenvalue prevents an exact diagonalization

For the supplied matrix, exact symbolic computation gives

$$\det(\lambda I - A) = (\lambda - 1)(\lambda - 2)^2.$$

Its eigenspaces are

$$E_1 = \text{Span}\{(1, -2, 2)^T\}, \qquad E_2 = \text{Span}\{(2, -5, 3)^T\}.$$

Each has dimension one, so there are only two independent eigenvectors for a three-dimensional matrix. Exact diagonalization is impossible. To see what happens to the requested checks, choose

$$D = \text{diag}(1, 2, 2), \quad P = \begin{bmatrix} 1 & 2 & 2 \\ -2 & -5 & -5 \\ 2 & 3 & 3 \end{bmatrix}.$$

Then $AP - PD = 0$, but $\det P = 0$ because columns 2 and 3 coincide. Thus PDP^{-1} is undefined. A zero eigenvector-equation residual alone does not make P an invertible change of basis. Floating-point eigenvalue commands may split the repeated root slightly and return nearly parallel eigenvectors. An attempted inverse then amplifies rounding errors; exact ranks resolve the issue. The following executed symbolic checks reproduce the conclusion:

```
from sympy import Matrix, eye, symbols, factor
A = Matrix([[-3,-2,0],[14,7,-1],[-6,-3,1]])
z = symbols('z')
print(factor(A.charpoly(z).as_expr()))
print((A-eye(3)).nullspace())
print((A-2*eye(3)).nullspace())
P = Matrix([[1,2,2],[-2,-5,-5],[2,3,3]])
D = Matrix.diag(1,2,2)
assert A*P-P*D == Matrix.zeros(3)
assert P.det() == 0
```

Exercise 5.S.26: A nonzero nilpotent matrix cannot be diagonalized

For the given 4×4 matrix, exact calculation gives

$$\det(\lambda I - A) = \lambda^4, \qquad \text{Nul } A = \text{Span}\{(1, 2, 1, 0)^T\}.$$

The only eigenvalue is zero, with algebraic multiplicity four but only one independent eigenvector. Thus the matrix is not diagonalizable. In fact $A^4 = 0$ and $A^3 \neq 0$, another exact check of its nonzero nilpotent structure. An eigenvector matrix for $D = 0_4$ can only have columns proportional to $(1, 2, 1, 0)^T$. Such a P has rank at most one: $AP - PD = 0$ is possible, but P^{-1} is impossible. Were an invertible P to exist, $PDP^{-1} = 0$ would contradict the given nonzero A. A floating-point eigenvalue routine can again produce small nonzero numbers around the repeated zero root and unreliable eigenvectors. The exact calculation that settles the issue is:

```
A = Matrix([[-8,5,-2,0],[-5,2,1,-2],
            [10,-8,6,-3],[3,-2,1,0]])
assert A.charpoly().as_expr().expand() == A.charpoly().gen**4
assert len(A.nullspace()) == 1
assert A**4 == Matrix.zeros(4)
assert A**3 != Matrix.zeros(4)
```

No external data are needed, and the failure is a proved mathematical obstruction, not a software or setup failure.